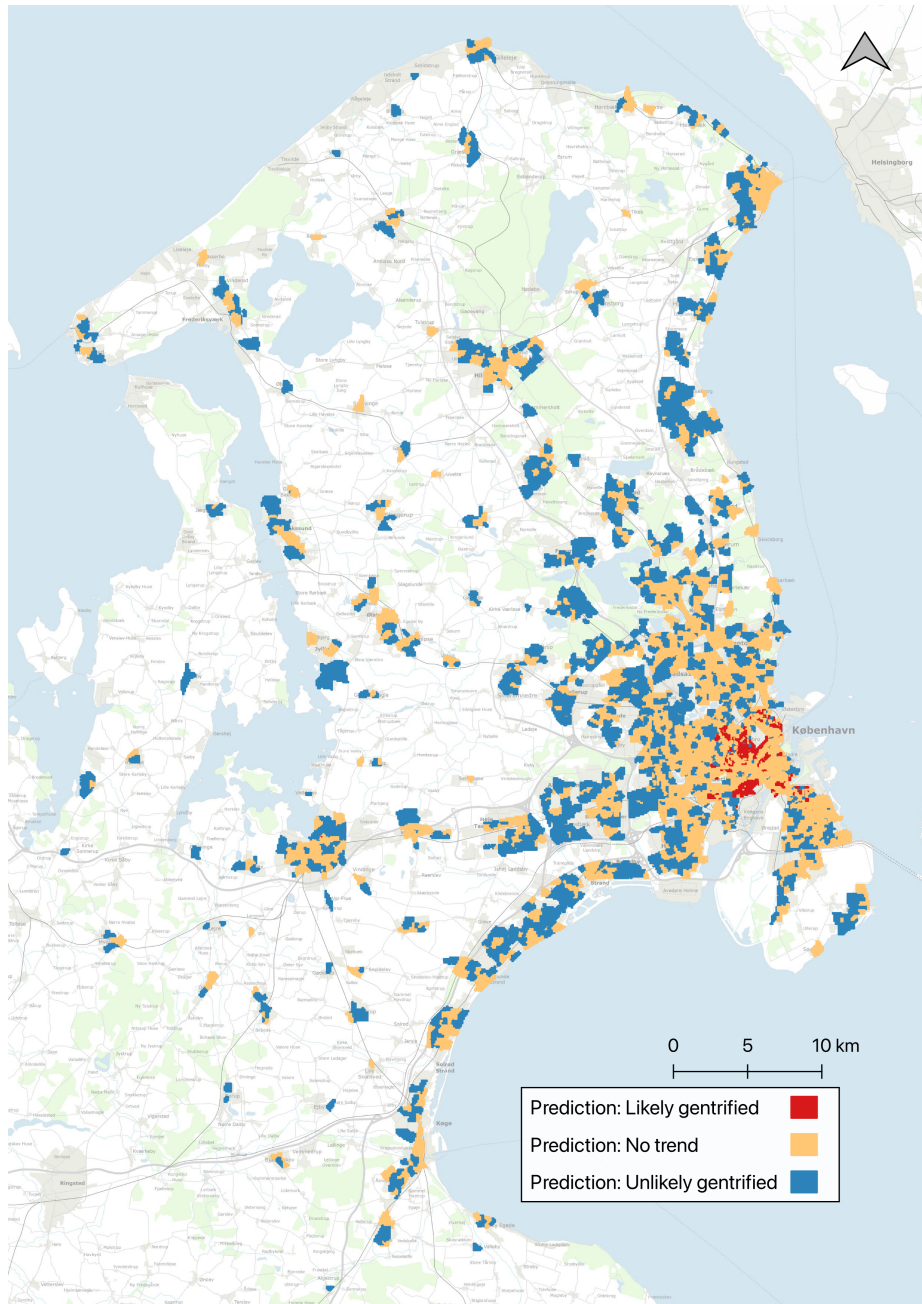


Predicting Gentrification with Machine Learning

Surveying, Planning and Land Management
1st semester, group 5, Autumn 2025



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Abstract:

Gentrification is a growing topic in the big cities that have seen a consistent population growth through the years. The ambiguous nature of gentrification and its consequences is derived from past research but it is unclear how it should be viewed in a Danish context. This study aims to predict areas prone to gentrification which is done by defining gentrification and its driving forces based on literature.

This knowledge is used to design a XG-Boost machine learning model based on nationwide socioeconomic data from The Danish Neighborhood Atlas. The final model predicts gentrified areas in 2020 with an overall accuracy of 79%. These predictions point to gentrification only occurring in bigger cities, where displacement is likely. The findings can provide city planners and policymakers with a tool to mitigate or counteract gentrification with displacement as a consequence. This can help make the big cities more equitable especially for marginalized and vulnerable communities.

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Synopsis in Danish

I takt med øget urbanisering i verdens storbyer har man set en markant stigning i efterspørgslen på boliger. Byernes attraktivitet forårsager mærkbare værdistigninger, hvilket kan føre til gentrificering. Gentrificering som term beskriver en udskiftning i befolkningssammensætning, som i praksis betyder, at den lavere sociale klasse bliver "presset" ud ad bydele til fordel for den højere sociale klasse. Denne tendens kan dokumenteres og kortlægges ud fra blandt andet socioøkonomiske befolkningsdata, hvilket er velbeskrevet i amerikansk, engelsk og australsk kontekst. I Danmark er der dog et videnshul som er værd at undersøge nærmere, hvilket har ført til udarbejdelsen af en problemformulering bestående af følgende arbejdsspørgsmål:

Hvordan kan gentrificering forstås i en dansk kontekst og er gentrificering det målbart?

I hvilken grad kan en machine learning model baseret på spatiale modeller, forudsige gentrificering i danske byer og bruges i en byplanlægningsproces?

Denne indledende undren besvares ud fra et litteraturstudie, hvor termet gentrificering og dets konsekvenser udspecificeres med udgangspunkt i eksisterende forskning. Dette gøres ved at se på fremtrædende definitioner, hvilke forudsætninger der fremmer gentrificering samt hvordan dette kan placeres i en dansk kontekst. Det undersøges også hvordan målbare socioøkonomiske data kan bruges til at identificere gentrificerede områder via machine learning (ML). Med henblik på at udvikle en ML model som kan anvendes i dansk kontekst, ses der nærmere på et udvalgt datasæt fra NabolagsAtlas fra perioden 1990-2020. Datasættet bygger på stedfæstet data fra Danmarks Statistik og dækker over Københavns Kommune, Frederiksberg Kommune og de resterende kommuner der fremgår indenfor Fingerplanens afgrænsning. På baggrund af gentrificeringstendenser bevist på Vesterbro og Nørrebro bliver data bearbejdet i ML modellen til at udpege områder med lignende mønstre. Udformningen af modellen baseres på XGBoost og yderligere optimering resulterer i en nøjagtighed på 79% til at identificere områder, der er tilbøjelige til at være gentrificeret.

Denne proces og dens resultater kortlægges og modellens anvendelighed diskuteres ud fra et planlægningsperspektiv. Her vurderes det, at modellen er brugbar som præventivt værktøj til byplanlæggere for at hæmme gentrificering og konsekvenserne derved. Slutvist konkluderes det, at ML modellen og dens udarbejdelse imødekommer problemformuleringens formål og der reflekteres yderligere over ML modellens forbedringspotentialer.

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PREFACE

PART I

INTRODUCTION

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PART VI

REFLECTION

A | Classifying Clusters in the Overlay Analysis

In this appendix the classified values of the time series clusters are explained. The values are used in the overlay analysis described in section ???. The time series clusters are visualized with graphs where time is shown on the horizontal axis as the number of years counting from 1990. By assessing the graphs, each time series cluster is assigned a value from 1 to 6 with the following categories:

1. Strong signs on being gentrified
2. Weak signs on being gentrified
3. Neither nor
4. Weak signs on not being gentrified
5. Moderate signs on not being gentrified
6. Strong signs on not being gentrified

This classification is based on the understanding from the literature described in chapter ???. The data is also compared to the national averages from DNA to understand if the development of the cluster is extraordinary or follows the national development. The social classes described in section ?? are also considered. The data from DNA includes the danish population aged 18-69 while the social classes only includes the population aged 18-59 and without students (Arbejderbevægelsens Erhvervsråd, 2021b).

Another factor affecting the data is that 17% of the population has moved from the two lowest social groups to the three highest groups in the 30 year period. This means that the three highest groups went from consisting of 24% of the population included in the social classes in 1990 to 41% in 2020 (calculated with data from (Arbejderbevægelsens Erhvervsråd, 2021a)). These conditions of the data have to be taken into account when comparing this data. The social classes are used when classifying clusters about education and labor market affiliation.

A.1 Family Type

In studies it is often assumed that the first demographic group to gentrify an area consists of young adults without children and with a higher levels of formal education (Cohen & Pettit, 2019). Couples without children have the biggest disposable income compared to couples with children and singles. Couples with children have a many expense connected to childcare while singles have a lower total income than couples (Bredahl et al., 2021, pp. 29 & 82). For gentrified areas it is thus expected that the percentage of couples without children will rise while the other two groups will decrease. Another possibility that has to been taken into account is that couples move into an area before having children, and then have a child and thus change groups. 2

Singles with and without Children

Figure A.1 display the percentage of residents in the area who are single with or without children. The national average of the percentage of singles went from 29% in 1990 to 34% in 2020 (Lange et al., 2023).

Average Time Series per Cluster - EMUB

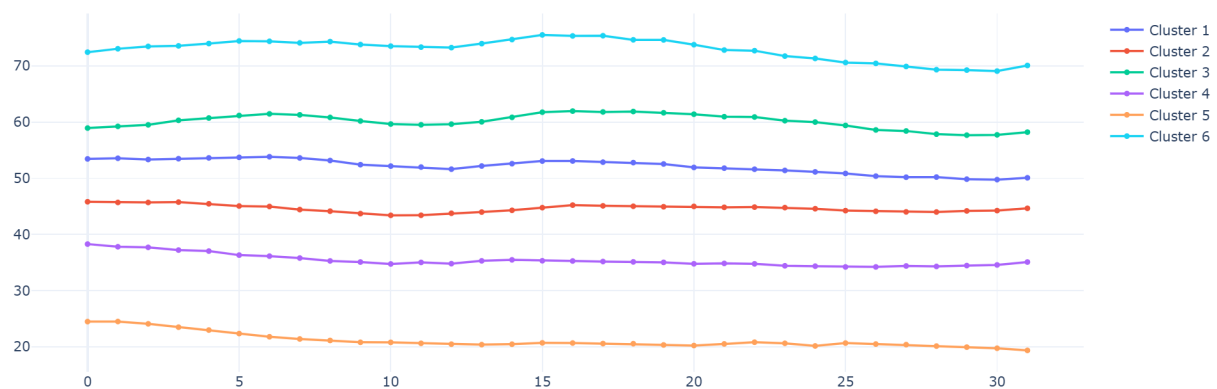


Figure A.1: Clusters of the dataset "singles with and without children". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents in the area who are single with or without children.

All the six clusters are more or less stable throughout the time period while the national average increase with 5%. The clusters shows limited signs on gentrification. Therefore the four top clusters are assigned the value 3 in table A.1. Cluster 4 and 5 are decreasing especially compared to the national average and they are also closest to the national average. Thus it is decided to categorize them with the value 2.

Table A.1: Assigned values for clusters of the dataset "Singles with and without children"

Singles with and without children						
Cluster	1	2	3	4	5	6
Value	3	3	3	2	2	3

Couples with Children

In the same way the time series cluster for the percentage of residents within the area who are couples or cohabiting with children is slowly increasing through the 30 year time period, presented in figure A.2. Thus all six clusters are assigned the value 3 as listed in table A.2. This is opposite the national tendency where the average drops from 45% in 1990 to 39% in 2020.

Average Time Series per Cluster - PMB

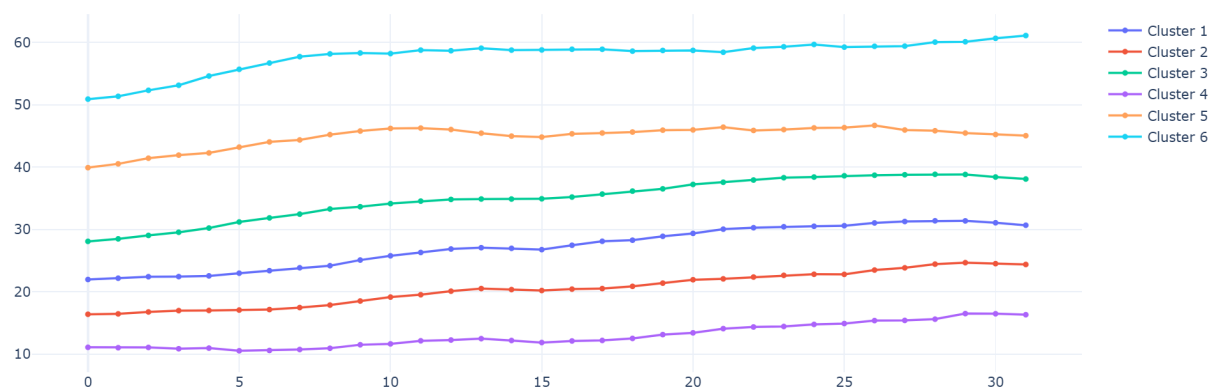


Figure A.2: Clusters of the dataset "Couples with children". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents within the area who are couples or cohabiting with children.

Table A.2: Assigned values for clusters of the dataset "couples with children"

Couples with children						
Cluster	1	2	3	4	5	6
Value	3	3	3	3	3	3

Couples without Children

The time series clusters for the percentage of residents within the area who are couples or cohabiting without children have a bigger variance than the previous two datasets. The

national average increase a little from 26% in 1990 to 28% in 2020.

Average Time Series per Cluster - PUB

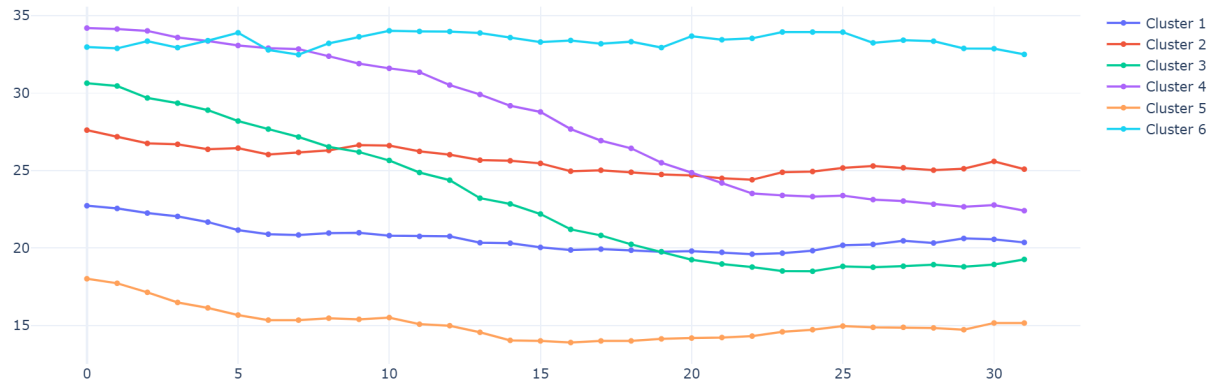


Figure A.3: Clusters of the dataset "Couples without children". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents within the area who are couples or cohabiting without children.

A cluster has to increase even further than the national average to make signs on gentrification, which none of the clusters in figure A.3 does. Cluster 3 and 4 are heavily decreasing, shown in figure A.3 and are thus assigned value 6, shown in table A.3. Cluster 2 and 6 are more stable and closer to the national average so they are classified with the value 3. Cluster 1 and 5 maintain their low percentage of couples without children throughout the whole period and they are thus valued 4.

Table A.3: Assigned values for clusters of the dataset "Couples without children"

Couples without children						
Cluster	1	2	3	4	5	6
Value	4	3	6	6	4	3

A.2 Age

As mentioned it is often younger adults without children and with higher levels of formal education who are the newcomers in gentrified areas (Cohen & Pettit, 2019). This fits the age group 26-40 years out of the age categories in the DNA data. The 18-25-year-olds are not finished studying or they do not choose a higher education and therefore they have a lower income. The two oldest categories consisting of the 41-69-year-olds can not be categorized as "younger adults". It is expected that for gentrified areas the amount of people aged 26-40 will rise while the other age groups will decrease.

Age: 18-25 years

The national average of residents aged 18-25 have increased with 2% from 17% in 1990 to 15% in 2020. In figure A.4 cluster 3 is decreasing the most with 8% but also cluster 1 is decreasing with 5%. They are thus assigned value 1 and 2, as shown in table A.4.

Average Time Series per Cluster - AGE_18_25

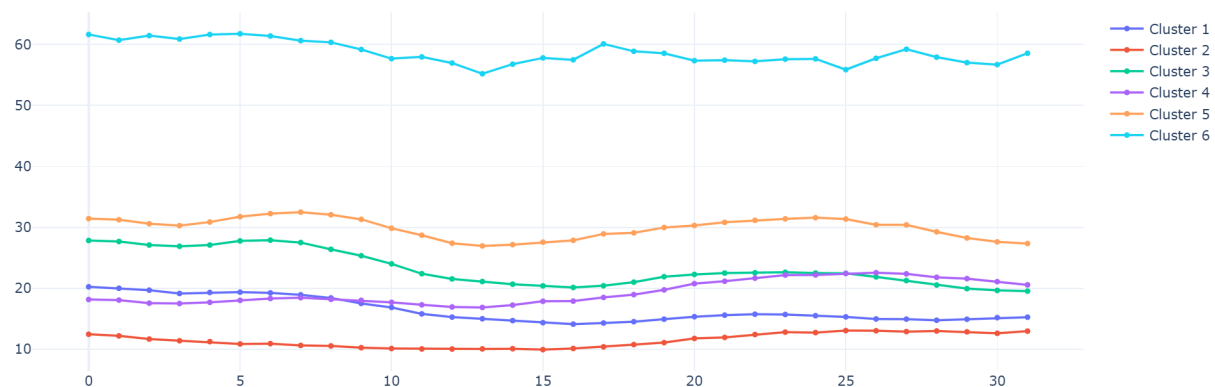


Figure A.4: Clusters of the dataset "Age: 18-25 years". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents aged 18-25 years within the area.

Cluster 4 is not changing significantly which is expressed in the value 3. The same applies for cluster 2 but the percentage is low and therefore the cluster is assigned the value 4. Cluster 5 is decreasing but is still 12% above the average of Denmark in 2020. Thus the cluster does not show signs on gentrification and is assigned the value 4. Cluster 6 is far above the other clusters and the national average and is thus classified as 6.

Table A.4: Assigned values for clusters of the dataset "Age: 18-25 years"

Age: 18 - 25 years						
Cluster	1	2	3	4	5	6
Value	2	4	1	3	4	6

Age: 26-40 years

For gentrified areas it is expected that the amount of residents in the age group 26-40 years will rise when the gentrification is taking place, because the new tenants are typically in this age group. After that the amount will lower, because the residents are staying to a greater extent and getting older and thereby shifting age group. In figure A.5 this tendency is present in both cluster 4 and 5 and they are thus given the value 2 and 1.

The national average of the percentage of 18-40-year-olds drops from 32% in 1990 to 28% in 2020. In 2002 the national average is 33% where cluster 4 is peaking with 57%.

Average Time Series per Cluster - AGE_26_40

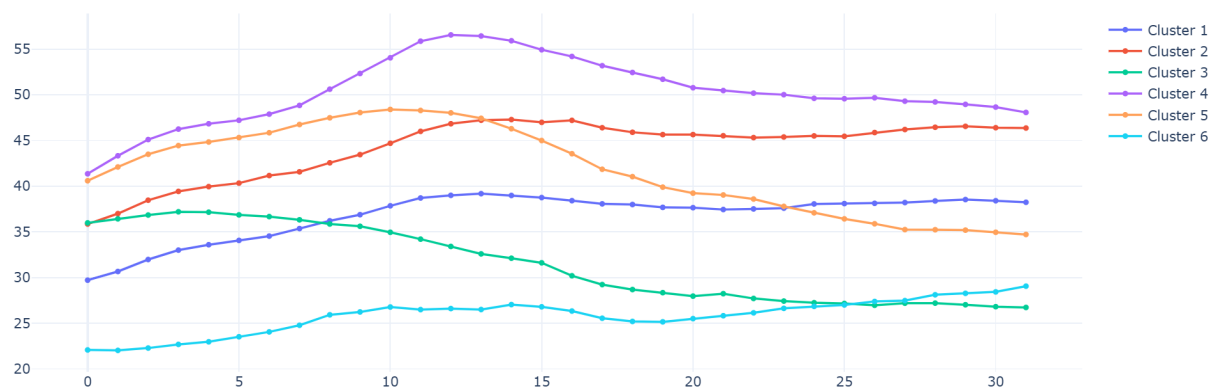


Figure A.5: Clusters of the dataset "Age 26-40: years". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents aged 26-40 years within the area.

Cluster 2 and 1 are not peaking as much as cluster 4, but they are still starting closer to the national average and then moving well above the national average. Therefore, they are valued 2. Cluster 6 is assigned the value 5 because it starts well below the national average, increases slightly and then ends just above. Cluster 3 decreases with 9% and therefore it does not show signs on gentrification and is classified with value 6.

Table A.5: Assigned values for clusters of the dataset "Age: 26-40"

Age: 26 - 40						
Cluster	1	2	3	4	5	6
Value	2	2	6	2	1	5

Age: 41-55 years

The age group of 41-55 years is more difficult to place than the two above. The percentage will lower when the age group gets displaced because of younger newcomers. It can also be argued that the percentage will gradually increase after the gentrification has taken place when the younger gentrifying group gradually transfers to the older age group. It is assumed that these conditions equalize each other for cluster 1 and 4, that both are decreasing just a little. The last argument is presumed to apply for cluster 2 and 3, because they are increasing after 2002. They are also assigned the value 2. None of them

are assigned the value 1 because the signs are not strong enough.

Average Time Series per Cluster - AGE_41_55

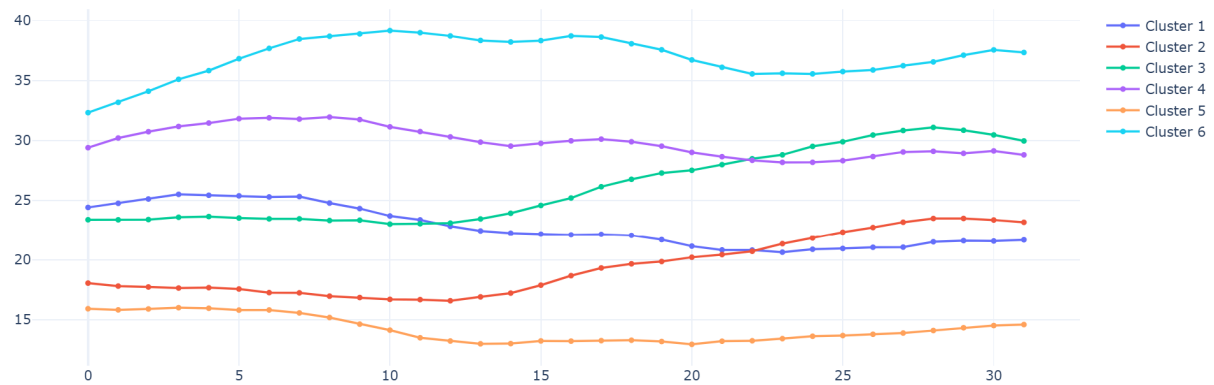


Figure A.6: Clusters of the dataset "Age: 41-55 years". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents aged 41-55 years within the area.

On national level the percentage of 41-55-year-olds is stable around 30% throughout the 30 year time period. Cluster 6 has a too big amount of this age group that indicates no gentrification. Cluster 5 is more neutral where gentrification cannot be extruded. It is thus classified with the value 3.

Table A.6: Assigned values for clusters of the dataset "Age: 41-55"

Age: 41 - 55						
Cluster	1	2	3	4	5	6
Value	2	1	1	2	3	6

Age: 56 - 69 years

The national level of 56-69-year olds has increased from 21% in 1990 to 27% in 2020. It is assumed that for areas that got gentrified in the period 1990-2020 the amount of 56-69 in 1990 is irrelevant but in 2020 it will be below the national average. This pattern is present for cluster 1, 2, 4 and 5 in figure A.7, that are all assigned the value 1. Cluster 5 increases above the national average and are assigned value 5. Cluster 6 is well above and therefore assigned the value 6.

Average Time Series per Cluster - AGE_56_69

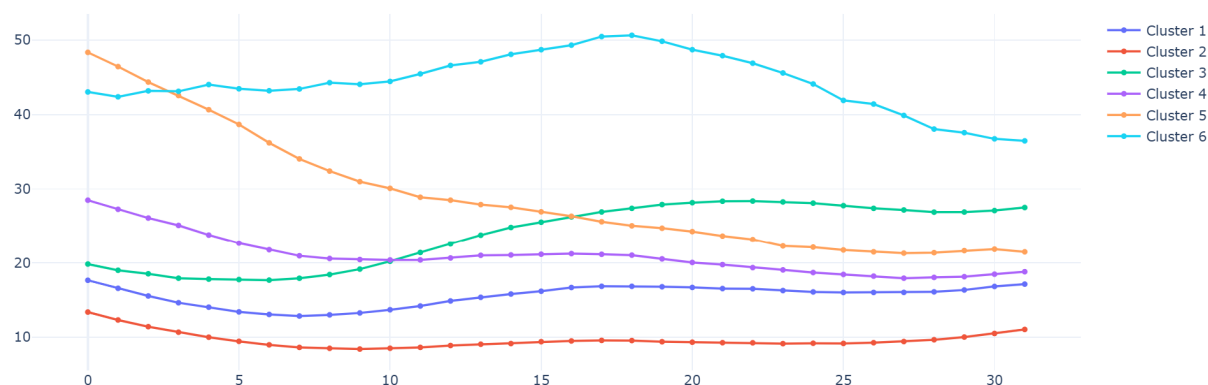


Figure A.7: Clusters of the dataset "Age: 56-69 years". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents aged 56-69 years within the area.

Table A.7: Assigned values for clusters of the dataset "Age: 59-69"

Age: 59 - 69						
Cluster	1	2	3	4	5	6
Value	1	1	5	1	1	6

A.3 Labor Market Affiliation

In gentrified areas it is expected to see the employment rate increasing while the unemployment rate and the amount of people outside the labor market will decrease (Zuk et al., 2015, pp. 29–30, Bredahl et al., 2021, p. 20)

Employed

The data consists of the percentage of employed residents aged 18-69 within the area. The rising employment rate applies to both cluster 1, 3 and 6, where all of them starts below the national average in 71% in 1990 and ends above the national average on 72% in 2020. This is illustrated in figure A.8. The development is most extreme for cluster 6, that is assigned the value 1 and cluster 1 and 3 is assigned the value 2.

Average Time Series per Cluster - EMP

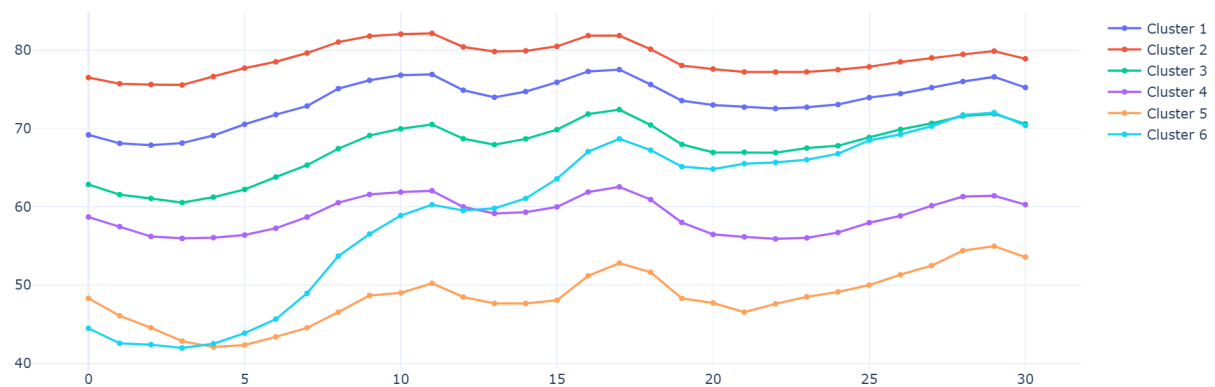


Figure A.8: Clusters of the dataset "Employed". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of employed residents aged 18-69 within the area.

Cluster 4 stays below the national average and is therefore assigned the value 4. Even more below is cluster 5 with the value 5. Cluster 2 has the highest percentage of employment and it therefore shows no signs of having been of lower class at any time. Cluster 2 does not appear to have been gentrified and is assigned a value of 5.

Table A.8: Assigned values for clusters of the dataset "Employed"

Employed						
Cluster	1	2	3	4	5	6
Value	2	5	2	4	5	1

Unemployed

The data consists of the percentage of unemployed residents aged 18-69 in the area. The graphs in figure A.9 shows that the unemployment rate dropped for most areas during the 1990's, but some more than others.

Average Time Series per Cluster - UNEMP

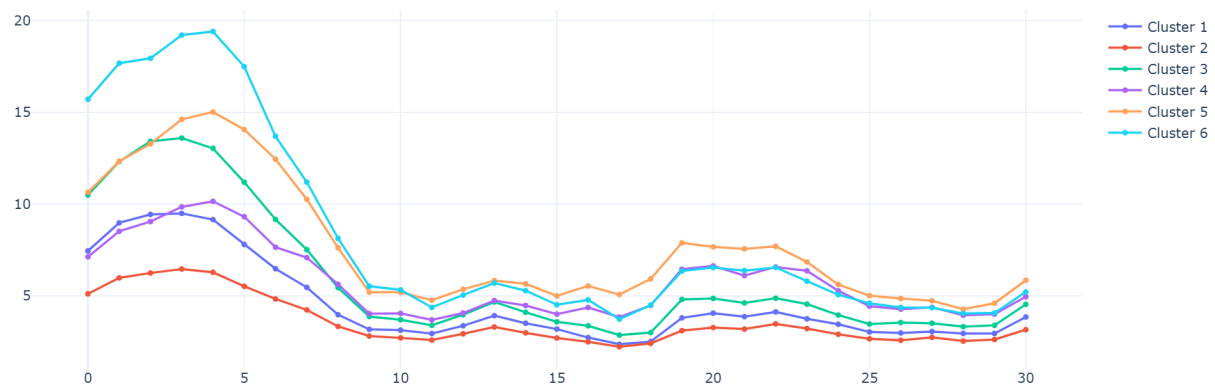


Figure A.9: Clusters of the dataset "Unemployed". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of unemployed residents aged 18-69 in the area.

The cluster with the biggest decrease is considered as showing most signs on gentrification, which is cluster 6. Below comes cluster 5 and then 3. Cluster 1 and 4 are very similar and the start values are close to the end values. Thus they are both classified with the value 5. Cluster 2 has the lowest percentage of unemployment and is thus assigned the value 6. Presumably it consists of prosperous residents.

Table A.9: Assigned values for clusters of the dataset "Unemployed"

Unemployed						
Cluster	1	2	3	4	5	6
Value	5	6	3	5	2	1

Outside the Labor Market

The data consist of the percentage of residents aged 18-69 who are currently outside the labor market. This includes retirees, students, and others not available for the labor market. When comparing to the social classes described in section ?? this data covers the social class called "long term unemployed" that are people who have been employed for less than 20% of the year and who are not self-employed. It can be early retirement pensioners, social security recipients and unemployment benefit recipients (Arbejderbevægelsens Erhvervsråd, 2020). Like the unemployment data, it is expected to see a decrease in this percentage for gentrified areas. Cluster 5 follows this pattern and goes down from 42% to 25%. But also cluster 2 starts above the national average on 22% in 1990 and ends below the national average on 25% in 2020.

Average Time Series per Cluster - OOL

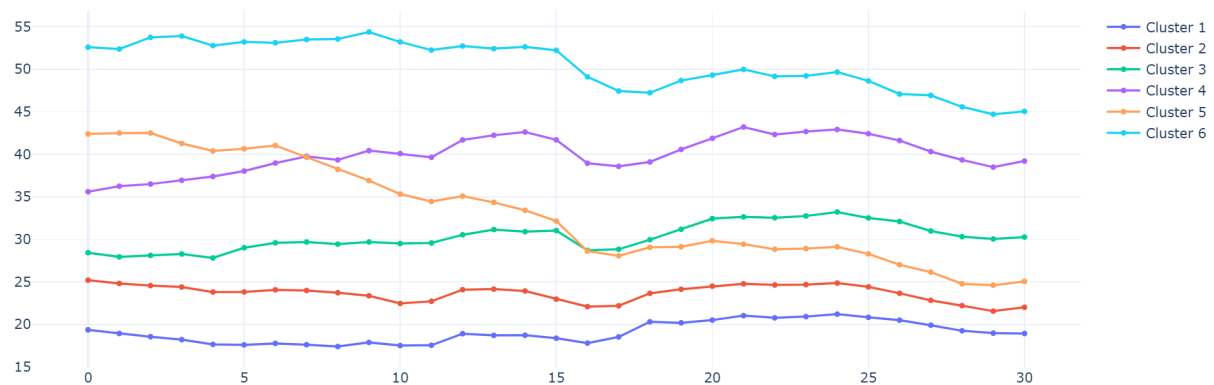


Figure A.10: Clusters of the dataset "Outside the labor market". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents aged 18-69 within the area who are currently outside the labor market.

Cluster 1 has a low percentage of residents outside the labor market throughout the whole period and is considered neutral with the value 3. Cluster 3 and 4 are rising and cluster 6 has a high percentage. This indicates that gentrification has not taken place in these areas and they are all three assigned the value 5.

Table A.10: Assigned values for clusters of the dataset "Outside the labor market"

Outside the labor market						
Cluster	1	2	3	4	5	6
Value	3	2	5	5	1	6

A.4 Education

As mentioned in section ?? the working class and the long-term unemployed are displaced from gentrified areas. This group of people typically has primary, secondary or vocational school as their highest level of education. Therefore the percentage of people with this education is expected to decrease for gentrified areas. The newcomers consists of people in the middle and upper class and they typically have a short- or long-cycle higher education. This percentage is thus expected to rise for gentrified areas. It also must be taken into account that the percentage of people with higher education as their highest education is rising at the national level while the percentage of people with vocational education as their highest education is diminishing (Statistics Denmark, n.d.). This means that the development of the cluster must be extraordinary from the development of the national

average.

Primary and Lower Secondary School

This data represents the percentage of residents who have primary and lower secondary school as their highest level of education. On national level this percentage has decreased from 42% in 1990 to 23% in 2020.

Average Time Series per Cluster - GRUND

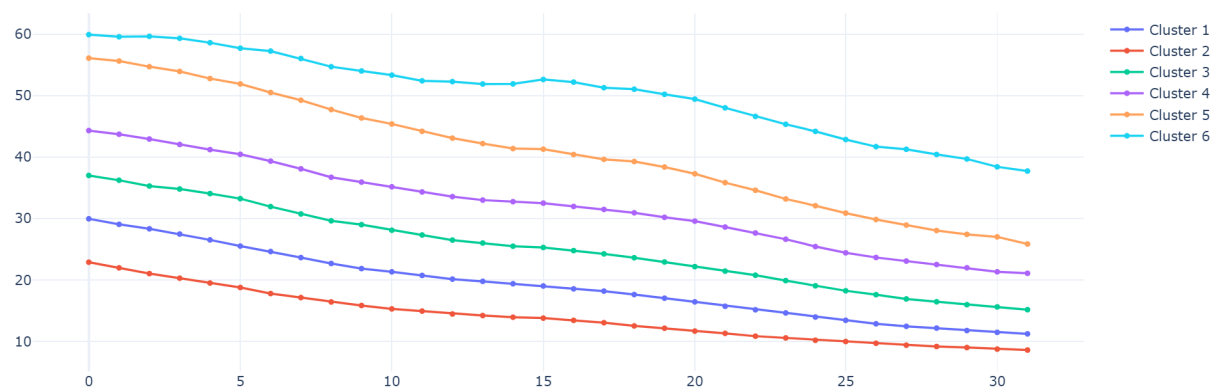


Figure A.11: Clusters of the dataset "Primary and lower secondary school". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents within the area who have primary and lower secondary school as their highest level of education.

Cluster 3 and 4 in figure A.11 are following the national development close but both ending just below the national average. This can indicate gentrification and they are assigned the value 1 in table A.11. Cluster 5 also drops in the percentage but still starts and ends above the national average. Cluster 1 only starts with a percentage in 30% but it still drops so it is assigned the value 2. Cluster 3 is considered neutral with the value 3 since it has the lowest level of change and it is well under the national average. Cluster 6 is assigned the value 5 since the amount never gets below or just close to the national average.

Table A.11: Assigned values for clusters of the dataset "Primary and lower secondary school"

Primary and lower secondary school						
Cluster	1	2	3	4	5	6
Value	2	3	1	1	2	5

Upper Secondary School and Vocational Colleges

This data consist of the percentage of residents within the area who have upper secondary school or vocational colleges as their highest level of education. On national level this percentage is 41% in 1990 to 42% in 2020. On figure A.12, The curve that follows the presumed gentrified tendency the most is cluster 2 and it is assigned the value 1. It starts well above the national average and ends well below. Cluster 1 and 3 are classified with the value 2, because they start just above the national average and ends below. This is also applies to cluster 4 but it ends on 37% in 2020 which is very close to the national average. Cluster 6 is well below the national average the whole period and is thus assigned the value 3. Cluster 5 is the only one rising and is thus assigned the value 6.

Average Time Series per Cluster - GYM_ERHV

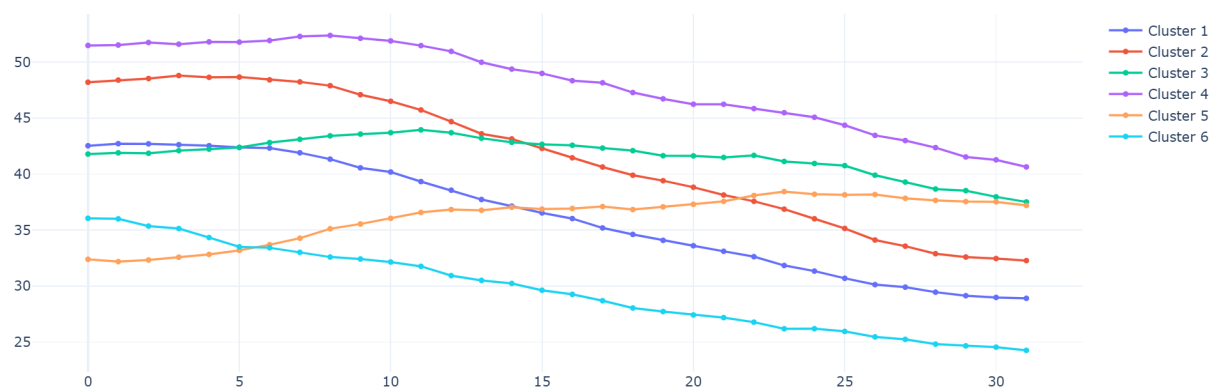


Figure A.12: Clusters of the dataset "Employed". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents within the area who have upper secondary school or vocational colleges as their highest level of education

Table A.12: Assigned values for clusters of the dataset "Upper secondary school and vocational colleges"

Upper secondary school and vocational colleges						
Cluster	1	2	3	4	5	6
Value	2	1	2	3	6	3

Higher Education

The data consists of the percentage of residents with higher education. The national average has increased from 16% in 1990 to 33% in 2020. Cluster 2 and 3 is close to the national average in 1990 and ends well above the national average in 2020. They are thus assigned the value 1. Cluster 1 also has a bigger difference on the starting and ending

point and starts at 22% in 1990 relatively close to the national average. Cluster 4, 5 and 6 is a little more flatten off and more outlying from the national average which is why they are assigned the values 5 and 6.

Average Time Series per Cluster - LVU

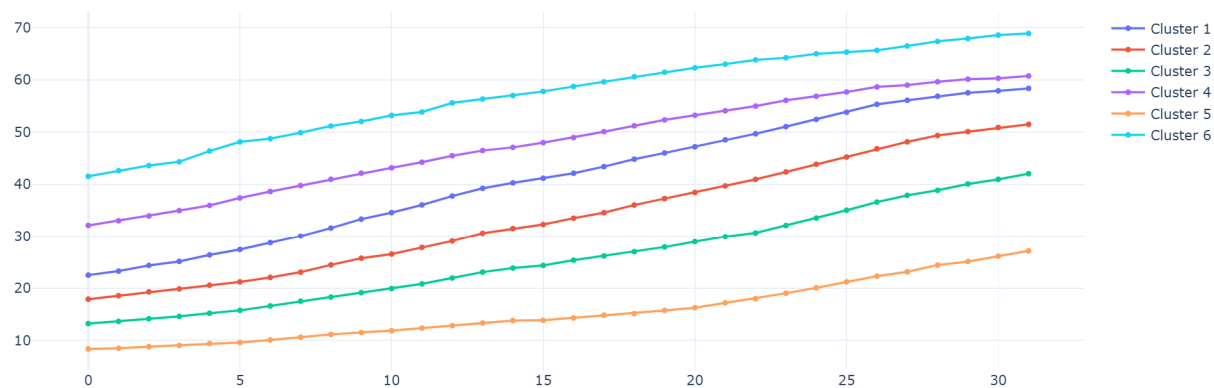


Figure A.13: Clusters of the dataset "Employed". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents within the area with higher education

Table A.13: Assigned values for clusters of the dataset "Higher education"

Higher education						
Cluster	1	2	3	4	5	6
Value	2	1	1	5	6	6

A.5 Residential Mobility

When gentrification is taking place there is a replacement of the existing residents with new residents according to the definition from section ???. That means that both the inflow and outflow of residents will rise in this period. After this when the area is gentrified the migration will assumably fade out. In general the in- and outflow is higher in Copenhagen and Frederiksberg Municipality than the rest of the country when comparing to figure A.14 and A.15. The national average of the in- and outflow was 4% in 1990 and 5% in 2020.

Inflow of Residents

On figure A.14 cluster 5 is sticking out because it peaks in the beginning and then flattens out. Therefore it is assigned the value 1. Cluster 2 is following the same pattern but to a lesser extent and is therefore assigned the value 2. It is possible that the neighborhoods in cluster 4 is more behind on the gentrification process and therefore they have only peaked and not flatten out. Thus cluster 4 is assigned the value 2.

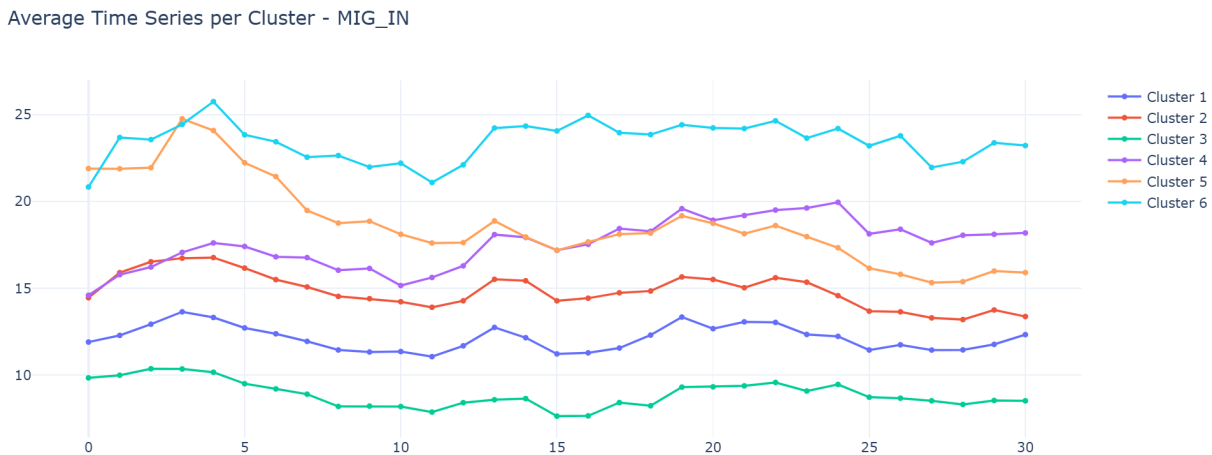


Figure A.14: Clusters of the dataset "Inflow of residents". Horizontal axis represents the number of years since 1990 and the vertical axis represents the outflow of residents from the area

Cluster 6 is more stable in the top and cluster 1 and 3 are more stable in the bottom. It does not indicate gentrification, which is why they are assigned the value 4.

Table A.14: Assigned values for clusters of the dataset "Inflow of residents"

Inflow of residents						
Cluster	1	2	3	4	5	6
Value	4	4	4	2	1	4

Outflow of Residents

The most significant peak of the curves in figure A.15 are the one by cluster 5 in 1994. This cluster is assigned value 1. The rest is more unclear but they are not showing signs on gentrification and are assigned the value 4. Cluster 6 is more jagged because it contains only a few neighborhoods. It is well above the other clusters and the national average and is thus classified with the value 6.

Average Time Series per Cluster - MIG_OUT

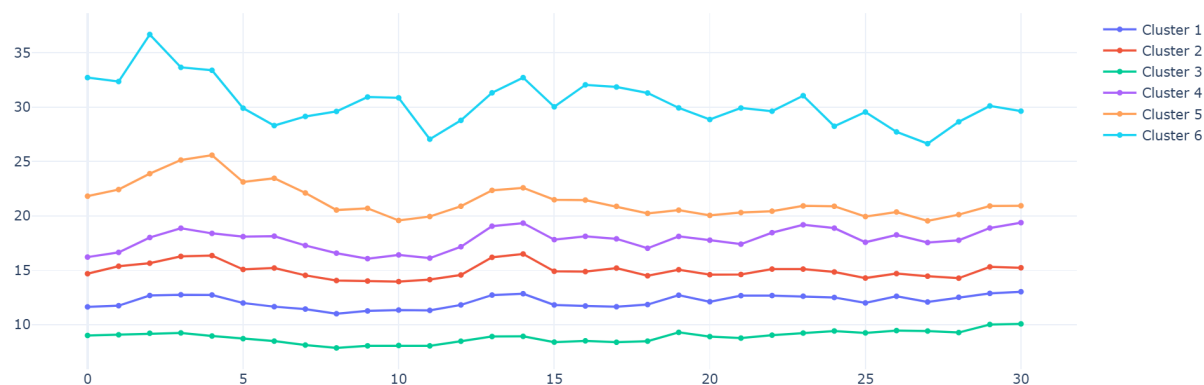


Figure A.15: Clusters of the dataset "outflow of residents". Horizontal axis represents the number of years since 1990 and the vertical axis represents the outflow of residents from the area

Table A.15: Assigned values for clusters of the dataset "Outflow of residents"

Outflow of residents						
Cluster	1	2	3	4	5	6
Value	4	4	4	4	1	4

A.6 Serious Crime

The data consist of the percentage of residents within the area who have been convicted of violating criminal laws, firearm or narcotics control laws. When an area becomes gentrified, this number will typically decrease. In figure A.16 especially cluster 5 but also cluster 4 are decreasing and they are thus assigned the values 1 and 2 in table A.16. Cluster 1, 2 and 3 are stable and are thus all assigned value 3. Cluster 6 only consists of one neighborhood around Sundholm at Amager, which is an institution for socially marginalized people. Cluster 6 is therefore unsuitable to point out signs on gentrification and is classified with the value 6.

Average Time Series per Cluster - CRIME_MAIN_Y

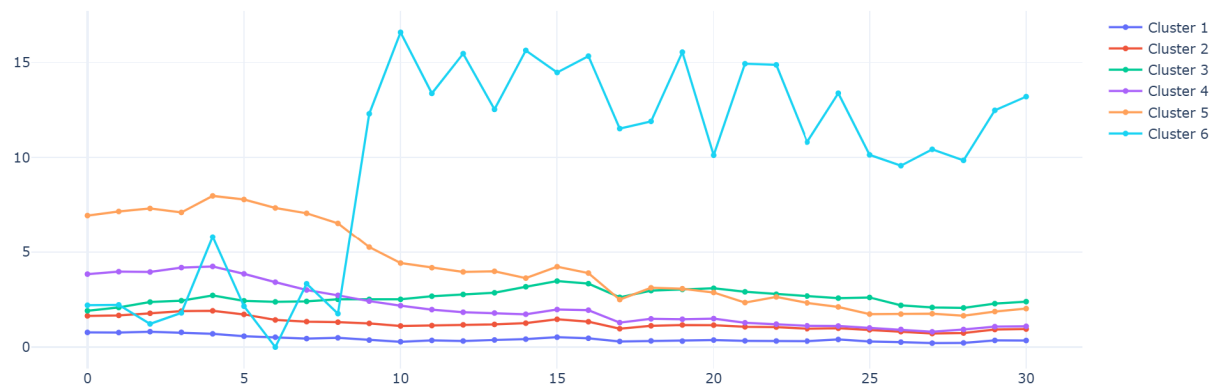


Figure A.16: Clusters of the dataset "Serious crime". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents within the area who have been convicted of violating criminal laws, firearm or narcotics control laws

Table A.16: Assigned values for clusters of the dataset "Serious crime"

Serious crime						
Cluster	1	2	3	4	5	6
Value	3	3	3	2	1	6

A.7 Disposable Income

The disposable income data is adjusted to the 2020 price level with the Net Price Index, which makes it possible to compare the data through the years without the impact of inflation. For gentrified areas the working class is replaced with the middle class or higher as described in section ??: *Social Classes*. In 1990 the national average of disposable income was 157,877 DKK while it was 248,014 DKK in 2020.

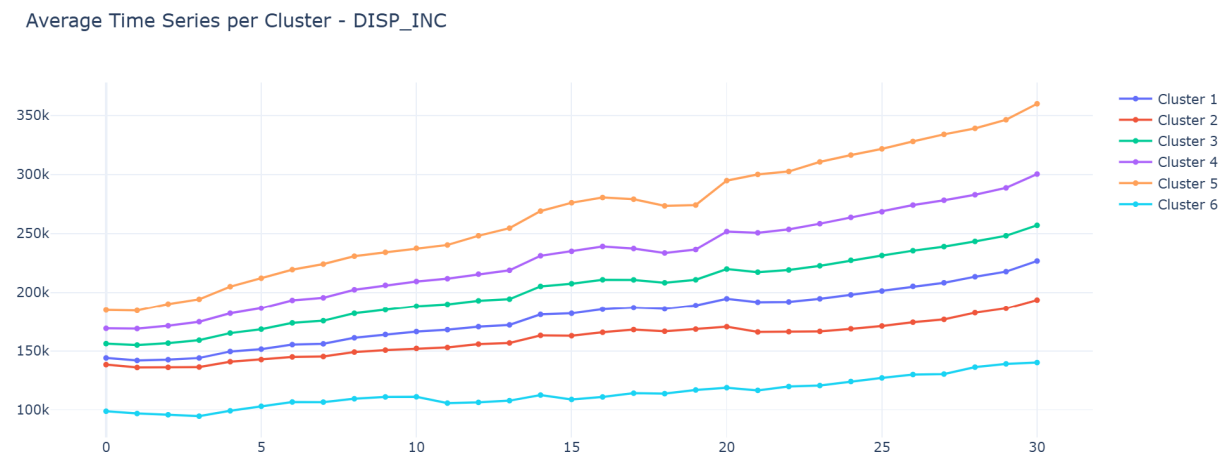


Figure A.17: Clusters of the dataset "Disposable income". Horizontal axis represents the number of years since 1990 and the vertical axis represents the average disposable income for residents in 2020 prices within the area

In figure A.17, Cluster 1 is below the national average in 1990 and ends close to the national average in 2020, which is causing the value of 1, shown in table A.17. Cluster 2 is similar but in 2020 the disposable income is a bit lower than for cluster 1, and therefore cluster 2 is assigned the value 2. Cluster 3 is above and it is thus more uncertain whether it has included as much working class as cluster 1, 2 and 6 in the early years. Therefore cluster 3 is assigned the value 3. The same applies to an even greater extent for cluster 4 and 5, whose income in 1990 is above the national average. Cluster 6 has only a small increase and is placed a certain amount below the other clusters. That indicates that more affluent inhabitants have not populated these areas included in cluster 6.

Table A.17: Assigned values for clusters of the dataset "Disposable income"

Disposable income						
Cluster	1	2	3	4	5	6
Value	1	2	3	4	5	6

A.8 Public Housing

When an area gets gentrified it is likely that the percentage of public housing in area have either stagnated or decreased. This is due to the gentrified neighborhood consisting mostly of private owned property which is renewable and therefor prone to follow the increasing house prices (Cohen & Pettit, 2019, p. 20). The national average of people living in public housing have been stable throughout the period from 18% in 1990 to 19%

in 2020. The curves in figure A.18 are all stable overall except cluster 6, that is rising drastically in 2008. This cluster is assigned the value 6. The three topmost clusters, cluster 3, 4 and 5, are all well above the national average and are thus also assigned the value 6. Cluster 1 and 2 are stable but with low concentration, which can indicate a larger sum of gentrifiable properties in those areas. These clusters are assigned the value of 3.

Average Time Series per Cluster - PUBLIC_HOUSING

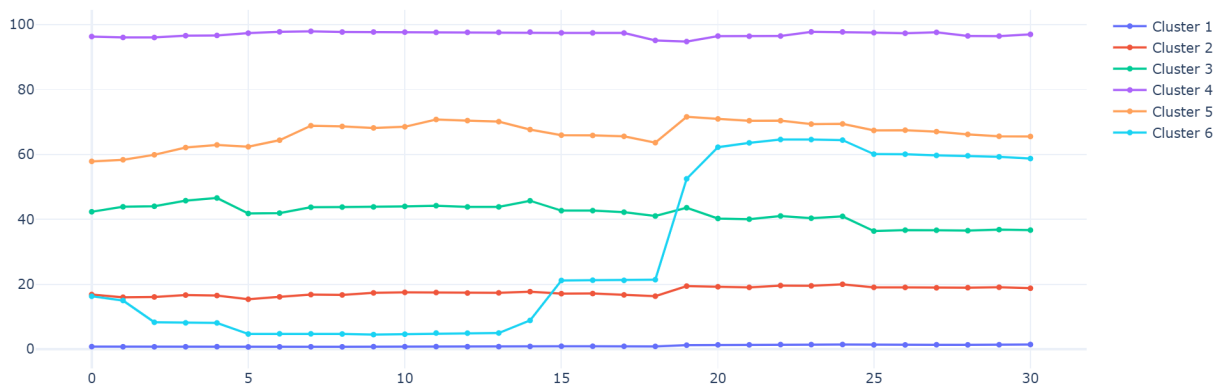


Figure A.18: Clusters of the dataset "Public housing". Horizontal axis represents the number of years since 1990 and the vertical axis represents the percentage of residents living in public housing within the area.

Cluster 1 and 2 are below the national average. They are classified as 3, because they do not show signs on being gentrified or not.

Table A.18: Assigned values for clusters of the dataset "Public housing"

Public housing						
Cluster	1	2	3	4	5	6
Value	3	3	6	6	6	6

A.9 Price per m²

The data consists of the average m² price for residential properties (houses and apartments). The prices are adjusted to 2020 prices with the Net Price Index. Notice that the curves in figure A.19 begin in 1992 two years later than the other cluster graphs due to lack of data.

Average Time Series per Cluster - MEAN_SQM

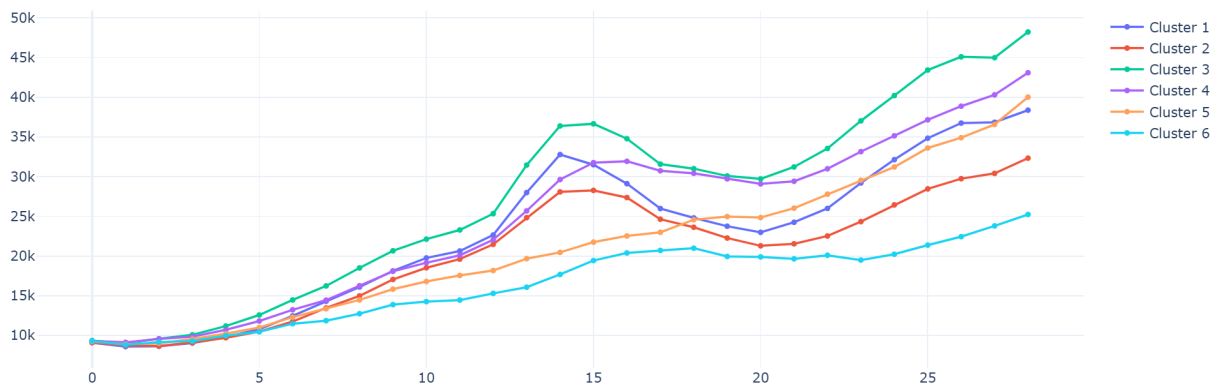


Figure A.19: Clusters of the dataset "Price per m^2 ". Horizontal axis represents the number of years since 1992 and the vertical axis represents the average m^2 price for residential properties (houses and apartments) in 2020 prices within the area [DKK]

As described in section ?? the housing price will rise in gentrified areas. In 1990 the national average of the price per m^2 was DKK 7,419 while in 2020 it was DKK 19,533. The clusters in figure A.19 are all above the national level but especially after the first ten years they become scattered. Cluster 1, 3, 4 and 5 are all assigned the value 1 because they are well above the national average. Then comes cluster 2 with the value 2. Cluster 6 is close to the national average and thus is assigned the value 3.

Table A.19: Assigned values for clusters of the dataset "Price per m^2 "

Price per m^2						
Cluster	1	2	3	4	5	6
Value	1	2	1	1	1	3